

Kshitij Bhardwaj

Third Year Undergraduate

Department of Mechanical Engineering

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📁 Portfolio | 🌐 Kshitij Bhardwaj | 🔗 kshitij-bhardwaj

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023-2027	B.Tech-Mechanical Engineering	Indian Institute of Technology Kanpur	8.0/10.0
2023	CBSE(XII)	Doon Public School, Dhanbad	96.6%
2021	CBSE(X)	De Nobili School, Dhanbad	97.2%

Scholastic Achievements

- Secured **AIR 3620** in **JEE Advanced** and **AIR 5850** in **JEE Mains**, outperforming over **11 lakh** candidates (2023)
- Secured **All India Rank 1638** in **KVPY SA** among the 50,000+ candidates, a rigorous research scholarship test (2022)

Major Competitions

FIFS Sports Big Data Gameathon | Inter IIT Tech Meet 13.0 - Dream11 MLOps Product Dev | 🌐 (Mar'25)

Problem	Automate fantasy team selection via MLOps pipeline maximizing fantasy points under real-world constraints
Approach	- Engineered 150+ contextual features (pitch, opposition, weather) via advanced scraping and ETL pipelines - Engineered sentiment-popularity signals on player name frequency (10d/30d), average and recent sentiments - Optimized for ODI, Test, T20 formats, dynamic training with iterative cascading and format-specific models
Results	- Achieved 19.9% MAPE, notably outperforming baseline models by >50% in fantasy point prediction accuracy - Secured 8th rank nationally in FIFS Gameathon and 6th among 23 IITs in the InterIIT Tech Meet 13.0

Key Projects

StyleGAN-Based Image Generation | EE655 Course Project | Prof. Koteswar Rao | 🌐 (Jan'25-Apr'25)

Objective	Style translation of unpaired images with semantic consistency and identity preservation across 2 domains
Approach	- Designed a CycleGAN architecture with ResNet blocks + with SE, CBAM, and Coordinate Attention - Curated custom dataset using Stable Diffusion (Ghibli-style), UTKFace with DeepFace-based preprocessing - Integrated SE-enhanced residual blocks and PatchGAN discriminators for fine-grained style translation - Optimized convergence with LSGAN loss, Adam optimizer, progressive scaling, and periodic visual validation
Results	Reduced generator loss from 1.75 to 1.05 and halved training convergence time using Coordinate Attention

Looking to Listen | Ksham Innovations - Startup | 🌐 (Aug'24-Dec'24)

Objective	Adapt Google's Looking to Listen pipeline for audio-visual speech separation and SDR benchmarking
Approach	- Retrained Google's Looking to Listen model in TensorFlow using audio-visual inputs and CUDA training - Automated dataset generation with yt-dlp, ffmpeg, MTCNN, and FaceNet for aligned face-audio extraction - Developed a custom benchmarks.py tool to compute SDR, SIR, and SAR for model performance analysis
Results	- Achieved SDR gain of +1.59 dB, with SIR of 4.24 dB and SAR of 8.94 dB for 2-speaker audio separation - Trained 100+ epochs with model accuracy of 70% and loss reduced from 0.9 to 0.3 on audio-visual validation

COCO-PUF Signal Race Prediction | CS771 Course Project | Prof. Purushottam Kar | 🌐 (May'24-Jul'24)

Objective	Predict dual responses of COCO-PUFs by modeling signal races across arbiter chains using linear classifiers
Approach	- Mapped 32-bit challenges to 1024-dim KR-product features enabling linear separability of responses - Trained SVM and Logistic Regression models with varying hyperparameters (loss, C, tol, penalty)
Results	Achieved 3.0%/0.7% error on secret data and 11.2s/37ms training/inference times under resource constraints

Voice Morph Companion | Electronics Club, IIT Kanpur (May'24-Jul'24)

Objective	Build a real-time voice assistant on a Single Board Computer (RPi-5) using speech, and chatbot models
Approach	- Trained a 3-class wake word detection model ("Noise", "Shazam", "Unknown") using Edge Impulse Studio - Deployed model on lightweight Arduino Nano BLE 33 Sense (256 KB SRAM) with ~400 ms inference delay - Integrated lightweight VOSK ASR for Indian English with 20+ user-recorded voice command augmentation
Results	Achieved 87.7% accuracy (quantized int8), outperforming 77.7% (float32 baseline), F1 0.89, ROC AUC 0.97

Technical Skills

- Programming Languages:** C, C++, L^AT_EX, Python, Java, Javascript
- Software and Libraries:** Git, Numpy, Pandas, Keras, Tensorflow, PyTorch, Scikit-Learn, Pratt, Matplotlib, React

Relevant Courses

(*:Ongoing)

Fundamentals of Computing-I (A) & II	Introduction to Machine Learning	Computer Vision and Deep Learning (A)
Signals, Systems and Networks*	Big Data Analysis through Copula	Linear Algebra and ODEs

Positions of Responsibility

Coordinator | Electronics Club, IIT Kanpur (Mar'25-Present)

- Leading a **two-tier** team structure comprising of **25** secretaries managing a budget of **1.8 Lakh** approved by Student Senate
- Mentored **ENIGMA** 🌀 (Cryptography, Embedded) and **GROOVE** (Signal Processing, ML) projects having 17 mentees.

BRAINS | Project Mentor | Stamatics, IIT Kanpur | 🌐 (May'25-Present)

- Mentored **50+** students on **Bayesian ranking** and **RL-tuned** investment strategy, with **uncertainty-aware** backtesting.